Lunar Settlement Vision: Autonomous Factory-First Approach

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January 4, 2026

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1 Introduction

Hello, my name is Jordan. We all want to explore the Moon, and there has been talk for decades about lunar bases. But now, more than ever, we have the technology — robotics, autonomous units, AI, and advanced materials — to build a sustainable settlement before humans even arrive.

Unlike other national or private plans, my vision is to first establish a self-sufficient industrial base on the Moon. This includes autonomous robots, 3D printers, and factories capable of using lunar resources to produce the infrastructure, energy, and life-support systems needed to support a small-scale base.

We begin by locating the base near confirmed ice deposits, but in sunlight nearly all the time to power solar arrays. Then, robots build the infrastructure to mine water, create reservoirs, and store essential materials such as oxygen, CO_2, and metals. Only after these resources are established do we construct a biodome for hydroponics, plants, and eventually humans.

2 The Plan

My proposal begins with a simple but critical shift in perspective: before we attempt to build a permanent presence on the Moon, we must first prove that we can do it autonomously on Earth.

I propose the creation of a research project or company—one that I would deeply value being part of—focused on developing a fully autonomous, AI-driven industrial factory capable of building all essential infrastructure *before* humans arrive at their destination. This factory would not be designed solely for space, but as a general-purpose autonomous industrial system that can operate in extreme, hostile, or inaccessible environments.

The first phase of this project would begin on Earth. We would construct a large-scale solar energy array capable of sustaining continuous industrial activity. Powered by this energy, we would then build an autonomous robotic factory—a single, integrated machine composed of modular systems—that can manufacture the core components required for off-world habitation.

This factory would be capable of:

- Smelting silicon and producing purified silicon crystals
- Manufacturing first-, second-, and third-generation solar panels
- Producing mirrors for thermal concentration systems (thermal candle arrays)
- Constructing water towers and sealed reservoirs
- Building electrolysis systems for hydrogen and oxygen production
- Fabricating robotic units, structural components, and replacement parts

Rather than viewing this as many separate machines, the factory should be understood as **one giant modular machine**—assembled incrementally, expanded through replication, and capable of building improved versions of itself.

A useful historical parallel can be found in ASML, the Dutch company responsible for developing extreme ultraviolet (EUV) lithography machines. For decades, scientists

believed such machines were physically impossible. The challenge required breakthroughs in optics, materials science, vacuum engineering, plasma physics, and precision manufacturing. Development took over 20 years, involved thousands of engineers and scientists, and required unprecedented international cooperation. Today, EUV lithography is indispensable to the production of the world's most advanced semiconductor chips.

What was once dismissed as impossible is now industrially routine.

The autonomous lunar factory proposed here should be approached with the same mindset. It is not a single invention, but a long-term systems engineering effort. If we can successfully build such a factory on Earth—capable of autonomous construction, repair, and expansion—then deploying it to the Moon becomes a problem of logistics rather than feasibility.

In practical terms, an initial lunar deployment might consist of a minimal seed system: a small number of autonomous rovers, a large deployable solar array, and a heavy-duty 3D printer capable of using local materials. This printer would fabricate modular components, robotic units, structural elements, and eventually additional printers and manufacturing tools. CNC-style subtractive manufacturing systems may supplement additive methods where higher precision or metalworking is required.

To ensure the technology is robust, the factory should first be built and tested in extreme conditions on Earth. Ideal locations include:

- Deserts such as New Mexico (USA), Botswana, or Namibia to simulate harsh, arid environments
- Estonia or other European countries for industrial infrastructure and access to research support
- Antarctica for cold and isolated testing to prepare for extreme lunar conditions

The funding and governance of such a project need not be limited to a single nation. This initiative could be supported by Estonia, the European Space Agency, NASA, the African Union, or a multinational consortium. The objective is not national prestige, but the development of a transformative industrial capability.

Importantly, the applications of this technology extend far beyond the Moon. The same autonomous factory could be deployed to Antarctica to construct research stations or entire settlements before human arrival. It could be used for disaster recovery, remote infrastructure development, or extreme-environment research. As with many foundational technologies, there are also clear defense and security implications, as autonomous construction systems could reduce human exposure in hazardous environments.

However, at its core, this proposal is not about militarization or competition. It is about proving that humanity can build sustainable infrastructure without relying on constant human presence. If we can master that capability on Earth, then establishing a permanent, self-sufficient presence on the Moon becomes not only achievable—but inevitable.

3 Known Lunar Resources

Current exploration and analysis have confirmed the presence of the following elements and compounds on the Moon:

- **Water Ice:** Definitive evidence shows that water ice exists in permanently shadowed regions (PSRs) near both the lunar north and south poles. These ice deposits have been detected using the Moon Mineralogy Mapper aboard Chandrayaan-1 and mapped by NASA's Lunar Reconnaissance Orbiter. Estimates from missions like Lunar Prospector suggest that the total amount of water ice could range from tens of millions up to billions of metric tons, though precise quantities and accessibility remain subjects of ongoing research. ([moon.nasa.gov](https://moon.nasa.gov/news/59/ice-confirmed-at-the-moons-poles/?utm$_source = chatgpt.com$))

- **Elements:** Hydrogen, Helium (including He-3), Oxygen, Silicon, Aluminum, Iron, Calcium, Magnesium, Titanium, Sodium, Potassium, Sulfur, Carbon.

- **Compounds / Minerals:** Regolith oxides such as SiO_2, Al_2O_3, FeO, MgO, CaO, TiO_2, and minerals like ilmenite ($FeTiO_3$), which contain oxygen bound in solid form.

Resources Not Confirmed / Missing

The Moon currently does **not** have confirmed deposits of the following elements and materials essential for building advanced electronics and life-support infrastructure:

- Phosphorus (essential for fertilizers, electronics, batteries)
- Boron (used in glass, ceramics, advanced alloys, and some solar technologies)
- Nitrogen (critical for breathable atmosphere and agriculture)
- Organic compounds (needed for soil creation, composting, and life support)
- Carbon (in sufficient quantities for long-term industrial use; only trace amounts detected)
- Some rare metals or dopants required for advanced electronics (e.g., lithium, cobalt, etc.)

Implications

Because these elements are not confirmed or abundant on the Moon, they would need to be **shipped from Earth** initially to support:

- Fertilizer for hydroponics and soil creation
- Electronics and solar panel dopants
- Batteries and other advanced industrial components
- Life-support systems requiring nitrogen or carbon-based compounds

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4 Known Human Technologies

Humanity currently possesses a variety of technologies relevant to building a self-sufficient lunar base. These include:

Autonomous Robotics

- **Mining robots:** Autonomous rovers capable of excavating regolith, ice, or minerals with limited supervision (e.g., NASA VIPER rover planned for the Moon).
- **Construction robots:** Large 3D-printing units that can print concrete or regolith-based structures autonomously on Earth; prototypes for extraterrestrial use are being tested.
- **Modular humanoid robots:** Robots that can assemble components, perform repairs, and manipulate objects in complex environments; examples include Boston Dynamics humanoid robots and NASA's Robonaut series.
- **Autonomous drones and transport units:** Capable of moving materials between mining sites and construction areas.

3D Printing and Manufacturing

- **3D printers for regolith-based construction:** Can create bricks, habitats, or structural components from lunar soil simulants.
- **Metal smelting and casting prototypes:** Experiments on Earth have produced small-scale methods for extracting iron, aluminum, and silicon from lunar regolith simulants.
- **Electronics and solar panel production:** Current methods exist for making silicon solar cells on Earth, but adapting them to lunar conditions is still in research phase.

Artificial Intelligence and Control Systems

- **AI for task scheduling and autonomy:** Robots can navigate, mine, and assemble structures with minimal supervision.
- **Environmental monitoring and life-support AI:** Current ISS systems allow semi-automated oxygen, water, and CO_2 management.

Technologies That Need Improvement

- Fully autonomous factories capable of building other robots, 3D printers, and modules entirely from lunar materials.
- Long-term operation of humanoid robots in lunar conditions (dust, extreme temperature swings, radiation).
- In-situ manufacturing of advanced electronics and solar panels using lunar materials (currently requires Earth-sourced dopants like phosphorus or boron).

- Closed-loop life-support systems fully operational before human arrival (oxygen, water, CO_2, and nutrient recycling).
- Autonomous construction of multi-story habitats or domes at lunar scale.

Summary

While current technology demonstrates that many robotic, AI, and construction tasks are feasible, there remain significant challenges for a fully autonomous, self-sufficient lunar base. My plan leverages these technologies while addressing their current limitations, focusing on resource pre-deployment, factory-first infrastructure, and pre-human ecosystem creation.

Existing Companies and Technologies in Context

While many aspects of my autonomous lunar industrial ecosystem have yet to be realized, there are companies and research projects that have developed key components. These technologies demonstrate that my proposal builds on real, operational systems, but no single entity currently integrates all of these capabilities into a fully autonomous, self-replicating factory ecosystem as envisioned here.

Autonomous Robotics:

- **Mining Robots:** NASA's VIPER rover, GITAI rovers, and Ceres Robotics systems can autonomously excavate lunar regolith and ice. These units demonstrate the feasibility of lunar mining, and in my plan, similar robots will be deployed in Phase 2 to extract water and other raw materials.
- **Construction Robots:** AI SpaceFactory and ICON prototypes illustrate 3D-printing habitats and structures using regolith simulants. These inform the modular 3D-printing factories in Phase 1 and Phase 5.
- **Humanoid and Modular Robots:** Boston Dynamics humanoids and NASA Robonauts can assemble, manipulate, and repair complex structures. My plan scales these capabilities for self-replicating factory assembly and infrastructure construction before human arrival.
- **Autonomous Transport Units:** Drones, robotic carriers, and rovers can move materials across difficult terrain. These are essential for moving regolith, water, and components between factory modules, reservoirs, and biodomes in Phases 0–5.

3D Printing and Manufacturing:

- **Additive Manufacturing:** AI SpaceFactory, ICON, and Earth-based 3D printing systems can produce structural elements and habitat modules. These technologies directly support Phase 1 autonomous factory construction and Phase 5 biodome assembly.
- **Metal Smelting and Casting Prototypes:** Small-scale experiments on Earth show extraction of iron, aluminum, and silicon from lunar regolith simulants. In my plan, scaled-up versions of these smelters enable the autonomous production of solar panels, structural components, and robot parts.

- **Electronics and Solar Panel Fabrication:** Silicon solar panels and electronics can be produced on Earth; lunar adaptation requires shipping dopants like phosphorus and boron. These systems are foundational for the Phase 0 solar array and Phase 1 factory.

Artificial Intelligence and Control Systems:

- **Task Scheduling AI:** Mission Control Space Services and PickNik Robotics provide AI for autonomous navigation, resource allocation, and construction task management. My plan scales this AI to orchestrate multi-robot factory operations across all phases.

- **Life-Support Monitoring AI:** Current ISS systems demonstrate semi-automated environmental control, which informs the pre-human closed-loop ecosystems in Phases 3–6.

Existing Gaps / Unique Features of My Proposal:

- No current company operates a **self-replicating, autonomous industrial factory** capable of mining, smelting, 3D-printing robots, and constructing infrastructure entirely autonomously.

- **Integrated resource storage systems:** My plan includes pre-deployment of reservoirs for metals, water, CO2, and other essentials, which is not currently implemented in any single project.

- **Pre-human ecosystem preparation:** Closed-loop hydroponics and atmospheric conditioning systems built before humans arrive are unique to this plan.

- **Phase-based autonomous expansion:** The concept of robots mining, building, and expanding infrastructure in discrete phases on the Moon without direct human intervention is novel.

Summary: By combining these already-existing technologies—autonomous robots, additive manufacturing, AI, and life-support monitoring systems—into a single modular, scalable, and autonomous industrial ecosystem, my proposal creates capabilities that currently do not exist. Existing companies provide proof-of-concept components, but only by integrating them into a *self-expanding lunar factory-first approach* can we achieve a pre-human, sustainable lunar base.

5 Comparative Overview of Lunar Settlement Plans

Feature	**CNSA (China)**	**NASA (Artemis)**	**SpaceX**	**Jordan Plan**
Robotics first?	Partial, robotic landers, sample return	Partial, some robotics pre-human	Mostly transport, some autonomous robotics	Fully autonomous factory before humans

Resource utilization	Ice mining for water	Ice mining, regolith 3D printing	Not detailed, future plans	Water, CO_2, regolith for metals, silicon, and factory materials
Factory / Industry	No	No, astronauts build infrastructure	Partial, human-built factories envisioned	Autonomous factory builds robots, 3D printers, furnaces, solar panels
Hydroponics / Plants	After humans arrive	After humans arrive	Conceptual for Mars, not detailed	Pre-human hydroponics in biodome, initial $CO_2 jumpstart via charco$
Animals / $CO_2 generation$	None	None	None	Small mammals or charcoal used to seed greenhouse gases pre-human arrival
Phased expansion / pods	Partial	Modular but astronaut-tended	Conceptual Mars colonization	Stepwise pods for humans, animals, crops, labs, bakeries, etc.
Closed-loop life support	Planned for humans	Planned	Not detailed	Fully autonomous pre-human closed-loop preparation
Unique innovation	Science-focused	Modular habitats	Transportation scale	Factory-first, autonomous ecosystem, pre-human CO_2, modular town-scale pods

6 Phased Lunar Base Development Plan

The development of a self-sufficient lunar base will follow a multi-phase approach. Each phase builds upon the previous, using autonomous systems, 3D printing, and lunar resources to establish infrastructure before humans arrive.

Phase 0: Solar Energy Infrastructure

Objectives: - Deploy large solar arrays and battery storage systems near confirmed ice deposits. - Provide continuous power for autonomous factory, mining, and life-support operations.

Autonomous Tasks: - Rovers install and orient solar panels for optimal sunlight exposure. - Connect battery banks, controllers, and communication units.

Materials to Ship from Earth: - High-efficiency solar panels - Batteries and energy storage modules - Power converters, controllers, and communication units - Autonomous installation rovers

Storage/Reservoir Considerations: - Battery banks for backup during lunar night cycles (if not fully on a sunlit peak)

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Phase 1: Autonomous Factory Construction

Objectives: - Build a factory capable of producing robots, 3D printers, modular components, and smelting metals from regolith. - Include furnaces for iron, steel, copper, and silicon production. - Factory can manufacture additional 3D printers, solar panels, and modular building components.

Autonomous Tasks: - Assembly of modular factory components - Setup of furnaces, smelting units, and printing stations - Manufacturing of initial robotic units for subsequent phases

Materials to Ship from Earth: - Core factory modules and robotic manipulators - Software control units - Furnace linings and initial catalysts for metal extraction - Tools for additive and subtractive manufacturing

Storage/Reservoir Considerations: - Dedicated storage for metals and lunar-mined regolith - Storage tanks for oxygen and hydrogen once electrolysis begins

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Phase 2: Ice Mining and Water Reservoir

Objectives: - Construct autonomous roads to ice deposits. - Robots mine ice and transport it to insulated water reservoirs. - Reservoir capacity scalable from tens to hundreds of cubic meters. - Water used for human consumption, hydroponics, and oxygen production via electrolysis.

Autonomous Tasks: - Excavation and transport of ice - Reservoir assembly and insulation - Pumping systems for water distribution

Materials to Ship from Earth: - Water reservoir components and insulation - Pumps, tubing, and valves - Initial tools for robotic maintenance

Storage/Reservoir Considerations: - Water reservoir large enough to support oxygen generation and initial hydroponics - Emergency water backup for humans upon arrival

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Phase 3: CO_2 Reservoir Setup

Objectives: - Generate initial carbon dioxide supply for plant photosynthesis. - Store in pressurized tanks to seed biodome atmosphere.

Autonomous Tasks: - Burn shipped charcoal to produce CO_2 (or manage small animal pods for biological CO_2 generation). - Fill and monitor $CO_2 tanks for pressure and stability.$

Materials to Ship from Earth: - Charcoal or organic material for CO_2 production - Pressurized $CO_2 tanks - Valves, sensors, and regulators$

Storage/Reservoir Considerations: - CO_2 reservoir sufficient to maintain photosynthesis and stable atmosphere in early hydroponics

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Phase 4: Resource and Auxiliary Reservoir Setup

Objectives: - Build reservoirs for essential lunar-mined materials: silicon, aluminum, iron, titanium. - Establish storage for materials humans will have to bring: nitrogen, phosphorus, boron, carbon-based compounds.

Autonomous Tasks: - Mining regolith for metals - Smelting and storing metals - Preparing materials for solar panels, construction, and life-support systems

Materials to Ship from Earth: - Nitrogen storage tanks - Phosphorus, boron, and carbon supplies - Containers for modular storage of metals and regolith-derived materials

Storage/Reservoir Considerations: - Each resource has dedicated tank or silo - Reservoirs designed for automated dispensing to robots and biodomes

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Phase 5: Biodome Construction

Objectives: - 3D-printed regolith dome with hydroponic walls interspersed between modular housing, labs, and office units. - Solar-powered environmental control. - Life-support pre-seeded with O_2 and CO_2.

Autonomous Tasks: - 3D printing of dome and structural components - Installation of hydroponic towers and life-support systems - Monitoring of oxygen, carbon dioxide, temperature, and humidity

Materials to Ship from Earth: - Hydroponic systems and nutrient solutions - Seeds, algae cultures, and starter plants - Monitoring and control equipment

Storage/Reservoir Considerations: - Reservoirs for water, CO_2, nutrients, and oxygen fully integrated into dome

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Phase 6: Planting and Hydroponics Activation

Objectives: - Robots plant initial crops and algae to begin oxygen production. - Automated nutrient, water, and CO_2 management starts stabilizing atmosphere.

Autonomous Tasks: - Seed planting, algae deployment - Continuous monitoring and adjustment of environment - Collection and recycling of water and nutrients

Materials to Ship from Earth: - Additional seeds and algae cultures as necessary - Supplementary nutrient and mineral packets

Storage/Reservoir Considerations: - Active nutrient circulation systems - Emergency reservoirs for O_2 and CO_2

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Phase 7: Human Arrival and Base Operation

Objectives: - Humans arrive after the ecosystem is stable. - Focus on compost, urine recycling, laboratory experiments, and base expansion. - Introduce additional food sources as pods are expanded.

Autonomous Tasks: - Continued maintenance of biodome, hydroponics, and life-support - Assistance for human habitat assembly and expansion

Materials to Ship from Earth: - Personal supplies, laboratory equipment, additional food - Specialized tools for human tasks

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Phase 8+: Modular Pod Expansion

Objectives: - Scale lunar settlement through animal pods (rabbits, goats, cows), crop pods (vegetables, fruits), and processing pods (bakery, butchery, labs). - Each pod functions as a self-contained ecosystem and production unit. - Allows for eventual town-scale lunar settlement.

Autonomous Tasks: - Construction of new domes and hydroponics - Automated feeding and care of animals - Waste management, composting, and nutrient recycling

Materials to Ship from Earth: - Initial livestock, seeds, and specialized equipment for food processing pods - Expansion materials for domes, hydroponics, and storage systems

Storage/Reservoir Considerations: - Separate reservoirs for each pod's water, nutrients, and atmospheric gases - Interconnected systems allow expansion without disrupting existing ecosystems

7 Conclusion

At first glance, this proposal may sound like something lifted straight from a science fiction novel. Perhaps I have been inspired a little too much by *Foundation*, *The Expanse*, and *Alien*, but the reality is that with modern technology, much of this plan is feasible.

The key insight is that we now have the tools to build autonomous industrial systems capable of constructing infrastructure without humans being present. Just as the Dutch company ASML spent decades—over 30 years and thousands of engineers—to develop extreme ultraviolet lithography machines once considered impossible, humanity today can build giant autonomous factories that can construct entire cities on the Moon. These machines are modular, self-expanding, and capable of mining, manufacturing, and assembling the resources required for life and industry.

The first step is testing this concept here on Earth. We can construct the factory in extreme and challenging environments such as New Mexico, Botswana, Namibia, Antarctica, or even Estonia. These regions simulate the harsh conditions of the Moon and allow us to validate autonomous operations: assembling solar arrays, 3D printing industrial components, building roads, mining raw materials, and filling reservoirs. By proving that such operations are possible on Earth, we eliminate many of the unknowns that come with lunar construction.

Once the autonomous systems are validated, we can leverage SpaceX's rocket infrastructure and other commercial launch providers to transport materials, robots, and initial resources to the Moon. From there, the factory can replicate itself, mine lunar resources, build water and CO_2 reservoirs, assemble biodomes, and cultivate hydroponic crops, all before humans arrive. By the time the first astronauts step onto the lunar surface, a sustainable, partially operational ecosystem and industrial base will already be in place.

This plan combines robotics, AI, additive manufacturing, energy independence, and life-support systems into a cohesive, phased approach. It demonstrates that humanity can

achieve self-sufficient off-world settlements without relying on continuous human labor from the outset. With vision, careful planning, and incremental testing on Earth, the Moon is not just a destination—it is a proving ground for humanity's next great industrial and ecological achievement.

My proposal: we start now, build the factory, begin testing in extreme Earth environments, validate autonomous systems, and then apply these lessons to the Moon. This approach will allow us to create a foundation for permanent human habitation that is efficient, scalable, and sustainable. The future of lunar cities begins not in orbit or on the Moon, but on Earth, where we can perfect the technologies, systems, and processes needed to survive—and thrive—on another world.